

IoT BASED SMART HOME SYSTEM WITH VOICE, BLUETOOTH, CLAP CONTROL AND AUTO PROTECTION

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AIM :

To design and implement an IoT-based smart home automation system using ESP32 that enables multi-mode control of electrical appliances through mobile app, voice commands, Bluetooth, and sound detection, with an added intelligent safety protection feature

APPARATUS REQUIRED :

S.No	Component	Qty	Purpose	
1	ESP32 Development Board	1	Main controller (processes all inputs & controls devices)	
2	3-Channel Relay Module	1	Switches Bulb1, Bulb2, and Fan safely	
3	LED Bulbs (230V)	2	Loads to demonstrate ON/OFF control	
4	Bulb Holders	2	Holds bulbs and connects to AC supply	
5	12V DC Fan	1	Additional load for automation	
6	12V Power Supply	1	Powers the DC fan	
7	Sound Sensor Module	1	Detects clap sound for control	
8	Jumper Wires	~20	Electrical connections between components	
9	Breadboard (optional)	1	Easy prototyping and connections	
10	USB Cable (ESP32)	1	Programming and power supply to ESP32	
11	Mobile Phone	1	Used for Blynk app control	

CORE CONCEPT OF THIS PROJECT

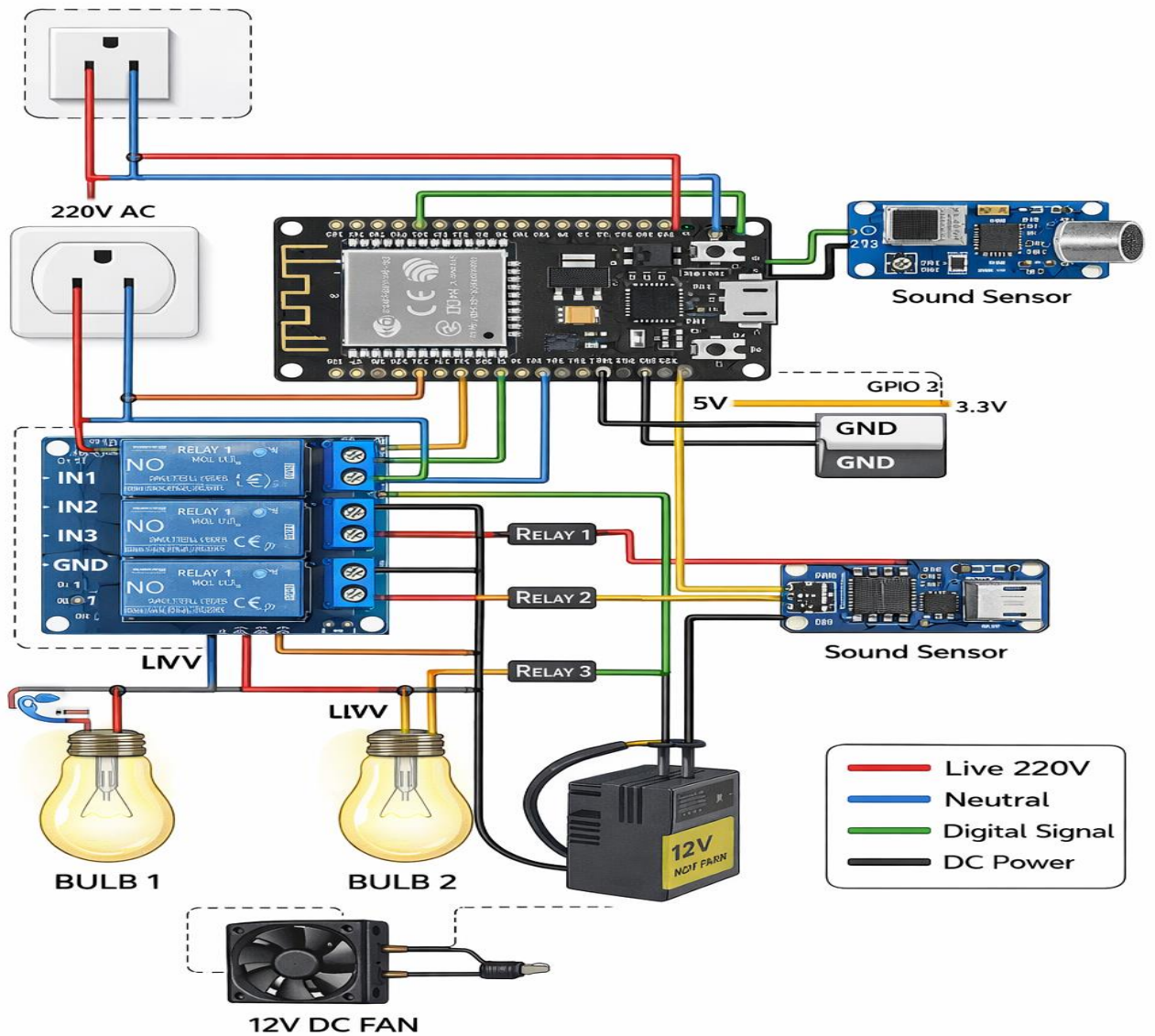
The core concept of this project is to develop a **smart home automation system** using the ESP32 microcontroller that enables control of electrical appliances through multiple methods such as mobile application, voice commands, Bluetooth communication, and sound (clap) detection.

The system integrates both **wired and wireless communication technologies** to provide flexible and user-friendly control over household devices like bulbs and fans. By using a relay module, low-power control signals from the ESP32 are used to safely switch high-voltage appliances.

Additionally, the project incorporates an **intelligent auto-protection feature**, which automatically turns off devices after prolonged usage to prevent energy wastage and improve safety.

Overall, this project demonstrates the combination of **IoT (Internet of Things), embedded systems, and smart control techniques** to create an efficient, reliable, and intelligent home automation solution.

CIRCUIT DIAGRAM

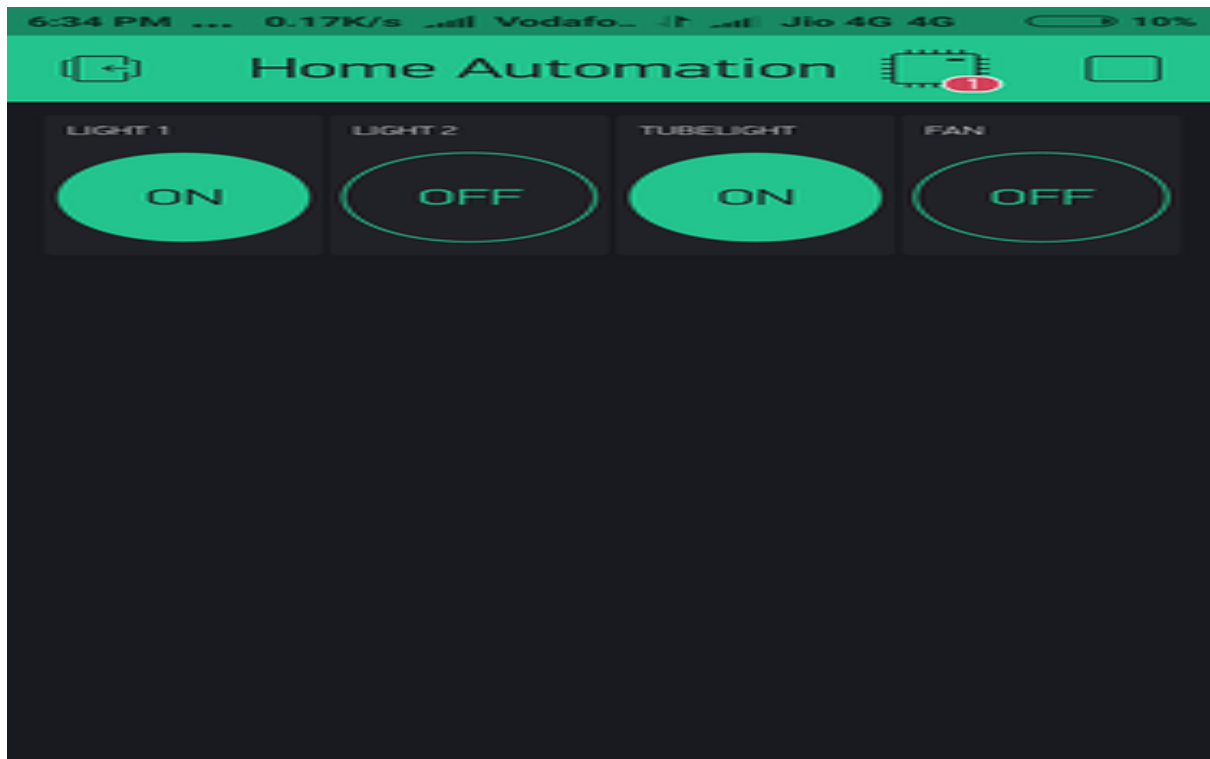
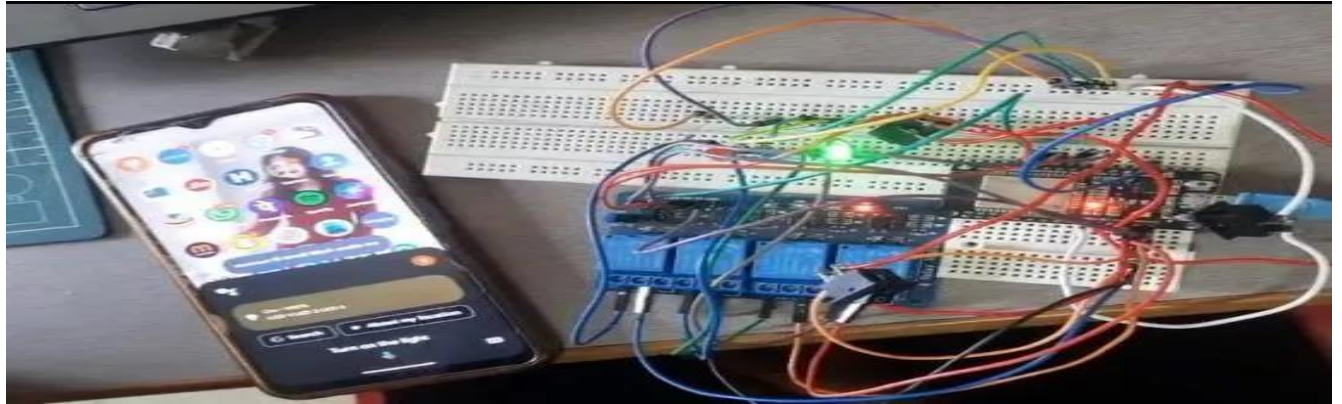


PROCEDURE :

Collect all components such as ESP32, relay module, bulbs, fan, sound sensor, and required power supplies.

1. **Connect the relay module to ESP32** using appropriate GPIO pins and provide 5V and GND connections.
2. **Wire the bulbs and fan through the relay module** ensuring proper connection of live and neutral lines.
3. **Connect the sound sensor to ESP32** and assign a digital input pin for clap detection.
4. **Install and configure the Blynk IoT app**, create datastreams (V0, V1, V2) and add switch widgets for device control.
5. **Write and upload the Arduino code** to ESP32 using Arduino IDE, including Blynk, Bluetooth, and control logic.
6. **Set up voice control using IFTTT**, linking Google Assistant commands to Blynk API for remote operation.
7. **Power ON the system and test all functionalities**, including app control, clap detection, Bluetooth operation, and automatic protection feature.

REAL TIME MODEL :



RESULT :

The smart home automation system was successfully implemented using ESP32. The appliances were controlled effectively through Blynk app, voice commands, Bluetooth, and clap detection with proper operation of all features.